# Yixuan Huang

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## Research Experience

**Postdoctoral Research Associate** with Prof. Tom Silver — *Princeton, NJ*
*Princeton University* — *Aug 2025 – Present*

**Visiting Student Researcher** with Prof. Jeannette Bohg — *Stanford, CA*
*Stanford University* — *Jan 2024 – Aug 2024*

**Graduate Research Assistant** with Prof. Tucker Hermans — *Salt Lake City, UT*
*University of Utah* — *May 2021 – Aug 2025*

**Undergraduate Research Assistant** with Prof. Sicun Gao — *La Jolla, CA*
*University of California, San Diego* — *Aug 2019 - Aug 2020*

## Education

**University of Utah, Salt Lake City, UT** — *Aug 2020 - Aug 2025*
*Ph.D. in Computing: Robotics*

- GPA: 3.98/4.0
- **Advisor:** Prof. Tucker Hermans

**Northeastern University, Shenyang, Liaoning (China)** — *Sep 2016 - Jun 2020*
*B.E. in Computer Science and Technology (**top** student in the department)*

- GPA: 93.2/100, Rank: 1/278

**University of California, San Diego, La Jolla, CA** — *Sep 2018 - Jun 2019*
*Exchange student*

- GPA: 3.91/4

## Publications

*[*]Equal contribution. [†]Student mentored by me.*

**Under Review**

[1] **Learning By and For Task and Motion Planning: A Survey**
*In Submission, 2026*
***Yixuan Huang**[*], Benned Hedegaard[*], Naman Shah, Siddharth Srivastava, George Konidaris, and Tom Silver*
[Paper]

[2] **TANDEM: Task and Motion Planning with As-Needed Demonstrations for Efficient Vision-Language-Action Model Fine-tuning**
*In Submission, 2026*
*Samrat Sahoo[*†], Liang Ji[*†], Tom Silver, and **Yixuan Huang***
[Project Website] [Paper]

[3] **DATAFARM: Distribution-Aligned Task and Motion Planning for Fine-Tuning Vision-Language-Action Models**
*In Submission, 2026*
*Samrat Sahoo[†], **Yixuan Huang**, and Tom Silver*
[Project Website] [Paper]

[4] **Coding Agents for Generalized Task and Motion Planning Problems**
*In Submission, 2026*
*Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, **Yixuan Huang**, and Tom Silver*
[Project Website] [Paper]

[5] **Chaining Visuomotor Skills for Constrained Manipulation Through Perceptual Manipulation**
*In Submission, 2026*
*Hao Wang†, **Yixuan Huang**, Somil Bansal, and Tom Silver*
[Project Website]

[6] **SPECTRE: Skeleton Priority Estimation Conditioned on TAMP Refinement Evidence**
*In Submission, 2026*
*Jiacheng Xu†, **Yixuan Huang**, and Tom Silver*
[Project Website]

[7] **Skeleton-Conditioned Base Placement Evaluation for Long-Horizon Mobile Manipulation**
*In Submission, 2026*
*Qihang Li†, **Yixuan Huang**, Tom Silver, and Peng Gao*

**Journal Publications**

[8] **Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers**
*IEEE Transactions on Robotics (T-RO), 2024*
***Yixuan Huang**, Nichols Crawford Taylor†, Adam Conkey, Weiyu Liu, and Tucker Hermans*
[Project Website] [Paper]

**Conference Publications**

[9] **KinDER: A Physical Reasoning Benchmark for Robot Learning and Planning**
*Robotics: Science and Systems (RSS), 2026*
***Yixuan Huang***, Bowen Li*, Vaibhav Saxena*, Yichao Liang, Utkarsh Aashu Mishra, Liang Ji†, Lihan Zha, Jimmy Wu, Nishanth Kumar, Sebastian Scherer, Danfei Xu, and Tom Silver*
[Project Website] [Paper]

[10] **RoboVista: Evaluating Vision Language Models for Diverse Robot Applications**
*Robotics: Science and Systems (RSS), 2026*
*Shuangyu Xie*, Kaiyuan Chen*, Ziyang Chen, Simeon Adebola, **Yixuan Huang**, Zehan Ma, Tianshuang Qiu, Wentao Yuan, Dhruv Shah, Pannag R Sanketi, and Ken Goldberg*
[Project Website] [Paper]

[11] **LAP: Language-Action Pre-training Enables Zero-Shot Cross-Embodiment Transfer**
*Robotics: Science and Systems (RSS), 2026*
*Lihan Zha, Asher James Hancock*, Mingtong Zhang*, Tenny Yin, **Yixuan Huang**, Dhruv Shah, Allen Z. Ren, and Anirudha Majumdar*
[Project Website] [Paper]

[12] **Hierarchical DLO Routing with Reinforcement Learning and In-Context Vision-Language Models**
*IEEE International Conference on Robotics and Automation (ICRA), 2026*
**Best Paper Award on Robot Learning – Finalist**
*Mingen Li, Houjian Yu, **Yixuan Huang**, Youngjin Hong, Hantao Ye, and Changhyun Choi*
[Project Website] [Paper]

[13] **Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference**
*Conference on Robot Learning (CoRL), 2025*
***Yixuan Huang**, Novella Alvina†, Mohanraj Devendran Shanthi, and Tucker Hermans*
[Project Website] [Paper]

[14] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
*IEEE International Conference on Robotics and Automation (ICRA), 2025*
***Yixuan Huang**, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg*
[Project Website] [Paper]

[15] **Out of Sight, Still in Mind: Reasoning and Planning about Unobserved Objects with Video Tracking Enabled Memory Models**
*IEEE International Conference on Robotics and Automation (ICRA), 2024*
***Yixuan Huang**, Jialin Yuan, Chanho Kim, Pupul Pradhan†, Bryan Chen, Li Fuxin, and Tucker Hermans*
[Project Website] [Paper]

[16] **Planning for Multi-Object Manipulation with Graph Neural Network Relational Classifiers**
*IEEE International Conference on Robotics and Automation (ICRA), 2023*
***Yixuan Huang**, Adam Conkey, and Tucker Hermans*
[Project Website] [Paper]

[17] **Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots**
*International Symposium on Medical Robotics (ISMR), 2021*
**Best Paper Award – Finalist & Best Student Paper Award – Finalist**
***Yixuan Huang**, Michael Bentley, Tucker Hermans, and Alan Kuntz*
[Paper]

## Workshop Papers

[1] **Learning Multimodal Probabilistic Models of Manipulation Skill Effects**
*CoRL 2025 Workshop on Learning to Simulate Robot Worlds*
*Siyeon Kim, Mohanraj Devendran Shanthi, **Yixuan Huang**, and Tucker Hermans*

[2] **Fail2Progress: Learning from Failures with Stein Variational Inference for Robot Manipulation**
*RSS 2025 Workshop on Out-of-Distribution Generalization in Robotics: Towards Reliable Learning-Based Autonomy*
**Best Paper Award – Finalist**
***Yixuan Huang**, Novella Alvina†, Mohanraj Devendran Shanthi, and Tucker Hermans*

[3] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
*CoRL 2024 Workshop on Learning Effective Abstractions for Planning (LEAP)*
**Oral Presentation**
***Yixuan Huang**, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg*

[4] **Latent Space Planning for Unobserved Objects with Environment-Aware Relational Classifiers**
*IROS 2023 Workshop on Causality for Robotics*
***Yixuan Huang**, Jialin Yuan, Weiyu Liu, Chanho Kim, Li Fuxin, and Tucker Hermans*

[5] **Reasoning and Planning about Unobserved Objects with Memory Models**
*CoRL 2023 Workshop on Neural Representation Learning for Robot Manipulation*
***Yixuan Huang***

## Honors and Awards

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| 2026 ICRA Best Paper Award on Robot Learning – Finalist | *May 2026* |
| Rising Star at the Princeton Robotics Inaugural Symposium | *April 2026* |
| 2025 ICRA Doctoral Consortium | *May 2025* |
| 2021 ISMR Best Paper Award – Finalist | *Nov 2021* |
| 2021 ISMR Best Student Paper Award – Finalist | *Nov 2021* |
| 2021 ISMR NSF Travel Award | *Oct 2021* |
| University of Utah School of Computing Department Fellowship | *Aug 2020* |
| National Scholarship (top 1%) | *Nov 2017 & 2018* |
| Northeastern University Excellent Student (top 1%) | *Dec 2017 & 2018* |
| Second Place in National Mathematical Modeling Competition in China | *Oct 2017* |
| First Place in Provincial Mathematical Modeling Competition | *Oct 2017* |

## Invited Talks

Sep 2026, "Towards Robots that Generate Micro-World Models On Demand", IROS 2026 RoBoWoMo Workshop [link], Pittsburgh, PA.

Sep 2026, "Planning Supervises Learning: Like, With, and Beyond Humans for Robot Foundation Models", Host: George Konidaris, Brown Robotics Seminar, Providence, RI.

June 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Rahul Mangharam, University of Pennsylvania, Philadelphia, PA.

Mar 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Felix Heide, Princeton University,

Princeton, NJ.

Feb 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Yashraj Narang, NVIDIA Research, Seattle, WA.

Sep 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Princeton Postdoctoral Council Seminar Series, Princeton, NJ.

July 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Oregon State University, Host: Li Fuxin, Corvallis, OR.

May 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", University of Texas at Austin, Host: Roberto Martín-Martín, Austin, TX.

Apr 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Robotics Seminar, Host: Haohan Zhang, Salt Lake City, UT.

Jan 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Graduate Research Symposium, Host: Pratik Soni, Salt Lake City, UT.

Feb 2024, "Reasoning and Planning for Unobserved Objects with Environment-Aware Relational Classifiers and Memory Models", University of Illinois Urbana-Champaign, Host: Katie Driggs-Campbell, Champaign, IL.

Nov 2021, "Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots", University of Utah Robotics Seminar, Host: Jake J Abbott, Salt Lake City, UT.

## Teaching

**ECE 531/COS 531: Robot Planning Meets Machine Learning** — *Princeton University*
*Guest Lecture about Learning Models for Robot Planning* — *Oct 2025*

**CS 6300 Artificial Intelligence** — *University of Utah*
*Guest Lecture about Reinforcement Learning* — *Nov 2024*

**CS 4300 Artificial Intelligence** — *University of Utah*
*Guest Lecture about Imitation Learning* — *April 2022*

**CS 4300 Artificial Intelligence** — *University of Utah*
*Teaching Assistant* — *Aug 2022 – Dec 2022*

**CS 4300 Artificial Intelligence** — *University of Utah*
*Teaching Assistant* — *Jan 2022 – May 2022*

## Mentoring

*Mentees who co-authored the publications listed above are marked with †.*

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|---|---|
| Samrat Sahoo† (Princeton Visitor → Stanford Master) | Summer 2026 |
| Hao Wang† (Princeton Visitor → Stanford PhD) | Summer 2026 |
| Liang Ji† (Princeton Visitor → Stanford Master) | Fall 2025 - Summer 2026 |
| Jiacheng (Joseph) Xu† (Princeton Master) | Fall 2025 - Present |
| Qihang Li† (NCSU PhD) | Spring 2026 - Present |
| Sabrina Liu (Princeton Master) | Fall 2026 - Present |
| Jingru Zhang (Princeton Visitor) | Fall 2026 - Present |
| Ian Henriques (Princeton Master → MIT PhD) | Fall 2025 - Spring 2026 |
| Tom Wang (Princeton Undergrad) | Fall 2025 |
| Akash Bhowmick (Princeton Undergrad) | Fall 2025 |
| Nichols Crawford Taylor† (Utah Undergrad → Northeastern PhD) | Fall 2021 - Spring 2024 |

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|---|---|
| Pupul Pradhan† (Utah Master → Evans & Sutherland) | Summer 2023 |
| Novella Alvina† (Utah Master) | Fall 2024 - Summer 2025 |
| John Chen (Utah Undergrad) | Spring 2025 - Summer 2025 |

## Workshop Organization Committees

**Learning Effective Abstractions for Planning (LEAP) @ CoRL 2026** — *Austin, TX*, *Oct 2026*
*Co-organizers: Naman Shah, Tom Silver, Utkarsh Mishra, David Paulius, Gregory Stein, Yiqing Xu, Amber Li, Franziska Herbert, Georgia Chalvatzaki*

## Reviewing

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|---|---|
| Conference on Robot Learning (CoRL) | 2023 - Present |
| Robotics: Science and Systems (RSS) | 2024 - Present |
| Robotics and Automation Letters (RA-L) | 2024 - Present |
| International Conference on Robotics and Automation (ICRA) | 2023 - Present |
| International Conference on Intelligent Robots and Systems (IROS) | 2024 - Present |
| International Conference on Learning Representations (ICLR) | 2025 - Present |
| Neural Information Processing Systems (NeurIPS) | 2026 - Present |
| IEEE Transactions on Robotics (T-RO) | 2025 - Present |
| IEEE Transactions on Robot Learning (T-RL) | 2026 - Present |
| The International Journal of Robotics Research (IJRR) | 2026 - Present |
| IEEE Transactions on Artificial Intelligence | 2024 |
| IEEE Transactions on Instrumentation and Measurement | 2024 |
| IEEE Transactions on Industrial Electronics | 2024 |
| Workshop on Learning Effective Abstractions for Planning (LEAP @ CoRL) | 2024 - Present |
| RSS Workshop on Out-of-Distribution Generalization in Robotics Towards Reliable Learning-Based Autonomy | 2025 |

## Outreach

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| Robotics lab tour co-organizer, University of Utah Bridge Program | 2022, 2023 |
| Introduction to mobile manipulator, Undergraduate AI Club at Princeton | 2025 |
| Introduction to robot programming for 7th-10th-grade students. | 2026 |

## Additional Service

**Princeton Robotics Seminar Organization Committee** — *Princeton, NJ*, *2025 - 2026*
*Co-organizers: Tom Silver, Jaime Fernández Fisac, Apurva Badithela, Merihan Alhafnawi, Kathryn Wantlin*